MOHAMMADREZA GOLKAR

AI Researcher | Machine Learning Engineer | PhD Candidate (AI Engineering)

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Research Interests

Explainable Artificial Intelligence (XAI) • Trustworthy AI • Fairness & Bias Analysis • Large Language Models (LLMs) • Foundation Models • Agentic AI • Multi-Agent Systems • Retrieval-Augmented Generation (RAG) • Medical Image Analysis • Computer Vision • Human-Centered AI

Professional Summary

M.Sc. graduate in Artificial Intelligence Engineering (GPA: 19.91/20, equivalent to 4.0/4.0 US GPA) with hands-on research experience in Explainable AI, Computer Vision, Large Language Models, and Agentic AI systems. Developed interpretable deep learning pipelines for medical image analysis using EfficientNet, Vision Transformers, Grad-CAM, and SHAP, achieving rigorous fairness evaluation across demographic subgroups. Built production-grade AI agent frameworks and RAG systems using LangChain, LangGraph, and FAISS. Over two years of industry software engineering experience. Seeking a PhD research position at Google to advance the frontiers of trustworthy, explainable, and human-centered AI systems.

Education

M.Sc. in Artificial Intelligence Engineering **2024 – 2026**

Islamic Azad University, Khomeini Shahr — Isfahan, Iran

- GPA: 19.91/20 (equivalent to 4.0/4.0 on the US scale) — ranked among the top graduates
- Thesis Grade: 18.5/20
- Thesis: *Fair and Explainable Deep Learning for Skin Lesion Classification* — developed interpretable deep learning models using EfficientNetB0 and Vision Transformers; applied Grad-CAM and SHAP for model transparency; conducted fairness and bias analysis across Fitzpatrick skin-type groups; evaluated with Accuracy, Precision, Recall, F1-score, and AUC

B.Sc. in Information Technology Engineering **2012 – 2017**

Islamic Azad University, Najafabad — Isfahan, Iran

Research Experience

Graduate Researcher — Thesis Project **2024 – 2026**

Islamic Azad University, Khomeini Shahr — Isfahan, Iran

Fair and Explainable Deep Learning for Skin Lesion Classification

- Fine-tuned EfficientNetB0 and Vision Transformer (ViT) models on a dataset of more than 10,000 dermatoscopic images, achieving high classification performance across 7 skin lesion categories
- Applied Grad-CAM and SHAP to produce saliency maps and feature importance explanations, improving model transparency for clinical decision support
- Designed and executed a fairness audit across 6 Fitzpatrick skin-type demographic groups, quantifying and mitigating performance disparities in minority classes

- Improved minority-class F1-score through targeted augmentation strategies including mixup, cutout, and class-weighted sampling
- Built end-to-end reproducible deep learning pipelines in PyTorch and TensorFlow, including data pre-processing, training, evaluation, and explainability modules

Projects

ResearchMatch-AI: Multi-Agent Research Position Matching System *2025*

- Designed and developed a multi-agent AI framework using LangGraph and LangChain to intelligently match researchers with PhD positions and research opportunities
- Built 5 specialized agents for CV analysis, skill extraction, research topic extraction, position analysis, and competency gap assessment
- Applied semantic similarity using Sentence Transformers and FAISS embedding-based retrieval over 1,000+ research position descriptions
- Generated personalized gap analysis reports and application strategy recommendations for each candidate-position pair
- Technologies: Python, LangGraph, LangChain, Sentence Transformers, FAISS, Streamlit, LLM Agents

Local Secure RAG Chatbot for Organizational Knowledge *2025*

- Built a fully local, privacy-preserving Retrieval-Augmented Generation (RAG) system for querying private organizational documents without any cloud data exposure
- Implemented a full pipeline covering document ingestion, chunking, embedding (Sentence Transformers), vector storage (FAISS), retrieval, and answer generation (Ollama)
- Reduced hallucination rate through source-grounded generation and confidence-based retrieval filtering
- Supported PDF, DOCX, and plain-text ingestion with hybrid dense/sparse retrieval strategies
- Technologies: Python, FAISS, Hugging Face Transformers, Ollama, LangChain

Semantic Search Chatbot *2025*

- Developed a semantic retrieval system over more than 10,000 text chunks using dense vector search (Sentence Transformers + FAISS), outperforming BM25 keyword-based baselines
- Built an interactive Gradio-based UI enabling real-time semantic query and ranked document retrieval
- Technologies: Python, FAISS, Sentence Transformers, Gradio

Multilingual PDF Summarizer *2024*

- Developed a local LLM pipeline for long-document summarization with automated multilingual translation using NLLB-200 models
- Automated document parsing, hierarchical chunking, abstractive summarization, and Persian/German/English translation workflows
- Technologies: PyTorch, Hugging Face, NLLB, NLP

LLM Interview Summarization Analysis *2024*

- Evaluated summarization behavior of FLAN-T5 and BART on real-world interview transcripts, comparing zero-shot and few-shot prompting strategies
- Analyzed factual consistency, coherence, and abstraction quality across prompting conditions
- Technologies: Hugging Face Transformers, Prompt Engineering, NLP

Industry Experience

Software Engineer — Mobile Applications **Sep 2023 – Sep 2024**
Megashid — Iran

- Developed and shipped Android applications using Kotlin and Flutter, serving thousands of active users
- Collaborated with cross-functional engineering teams on Industry 4.0 and IoT product integrations
- Improved app performance by refactoring memory-intensive modules, reducing crash rate and improving responsiveness

Software Engineer — Mobile Applications **Mar 2022 – Aug 2023**
Terapico — Iran

- Developed Android applications for digital healthcare platforms serving end users in Iran
- Implemented scalable mobile features following MVVM architecture and integrated REST APIs and Firebase backend services
- Collaborated with product and engineering teams across full development lifecycle (design, development, QA, release)

Technical Skills

Languages	Python, SQL, Kotlin, Bash, Git, Linux
ML Frameworks	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
LLM & Agents	LangChain, LangGraph, Ollama, OpenAI API, Prompt Engineering
Explainable AI	Grad-CAM, SHAP, Fairness Analysis, Bias Assessment, Saliency Maps
Computer Vision	EfficientNet, Vision Transformers (ViT), Image Classification, Medical Imaging
RAG & Retrieval	FAISS, Sentence Transformers, Embedding Models, Vector Databases
Software Eng.	REST APIs, Firebase, MVVM, Android (Kotlin/Flutter), n8n Workflow Automation
Tools	Gradio, Streamlit, Jupyter, Docker (basic), Weights & Biases

Languages

English — TOEFL iBT 4.5/6.0 (B2–C1, equivalent ≈90/120) **German** — B2 (Upper Intermediate)
Persian — Native

Key Achievements

- Achieved M.Sc. GPA of 19.91/20 (4.0/4.0 equivalent), among the highest in the program
- Thesis awarded a grade of 18.5/20 for contributions to fair and explainable medical AI
- Designed and deployed 5+ end-to-end AI systems spanning XAI, multi-agent frameworks, and RAG pipelines
- Over 2 years of industry software engineering experience shipping production mobile applications
- Open to PhD research opportunities in Explainable AI, Foundation Models, Agentic AI, Trustworthy ML, and Medical AI